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Short Communication

Inhibition of nociceptive dural input to the trigeminocervical complex through oxytocinergic transmission

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ABSTRACT

Migraine is a complex brain disorder that involves abnormal activation of the trigeminocervical complex (TCC). Since an increase of oxytocin concentration has been found in cerebrospinal fluid in migrainous patients and intranasal oxytocin seems to relieve migrainous pain, some studies suggest that the hypothalamic neuropeptide oxytocin may play a role in migraine pathophysiology. However, it remains unknown whether oxytocin can interact with the trigeminovascular system at TCC level. The present study was designed to test the above hypothesis in a well-established electrophysiological model of migraine. Using anesthetized rats, we evaluated the effect of oxytocin on TCC neuronal activity in response to dural nociceptive trigeminovascular activation. We found that spinal oxytocin significantly reduced TCC neuronal firing evoked by meningeal electrical stimulation. Furthermore, pretreatment with L-368,899 (a selective oxytocin receptor antagonist, OTR) abolished the oxytocin-induced inhibition of trigeminovascular neuronal responses. This study provides the first direct evidence that oxytocin, probably by OTR activation at TCC level inhibited dural nociceptive-evoked action potential in this complex. Thus, targeting OTR at TCC could represent a new avenue to treat migraine.

1. Introduction

Migraine is a common and disabling neurological disorder involving a dysfunction of the trigeminovascular system (Goadsby et al., 2017). During migraine, nociceptive trigeminal sensory nerves mediate excitatory neurotransmission through the trigeminocervical complex (TCC). Molecules that inhibit nociceptive transmission at this level could be considered as potential antimigraine agents (Akerman et al., 2013). Recently it has been reported that oxytocin (a hypothalamic neuropeptide) seems to relieve migraine pain (Tzabazis et al., 2016, 2017; Wang et al., 2013; You et al., 2017). Nevertheless, how and where this neuropeptide is able to exert its antimigraine action remains uncertain.

At spinal lumbar level, it has been shown that oxytocin specifically inhibits the nociceptive input arriving from the periphery. Briefly, electrophysiological recordings in anesthetized rats of spinal convergent second order wide dynamic range (WDR) neurons showed that oxytocin inhibited the neuronal firing associated with the activation of Aδ- and C-fibers nociceptors, but not proprioceptive Aβ-fibers (for refs. see Gonzalez-Hernandez and Charlet, 2018), pointing to its preferential blocking of nociceptive transmission. Similar results were recently reported at the level of the medullary dorsal horn (Sp5C-C1) where oxytocin seems to act on oxytocin receptors (OTRs) to inhibits nociceptive transmission from the ophthalmic dermatome (García-Boll et al., 2018).

On this basis, the present study set out to investigate in a well-established migraine rat model of acute dural nociceptive activation, the effect of oxytocin on TCC neuronal transmission in response to dural trigeminovascular activation. Since this model is considered predictive of potential antimigraine compounds (Akerman et al., 2013), we present evidence for a specific role of oxytocin in modulating the dural nociceptive input at the TCC by OTR activation.

2. Material and methods

2.1. Animals and husbandry

All procedures were approved by the local animal welfare committee and in accordance with NIH (NIH Publications No. 8023, rev. 1978) and ARRIVE guidelines. A total of 18 adult Wistar rats (290–320 g) provided by our bioterium were used. The rats with food and water *ad libitum* were individually housed in plastic cages with

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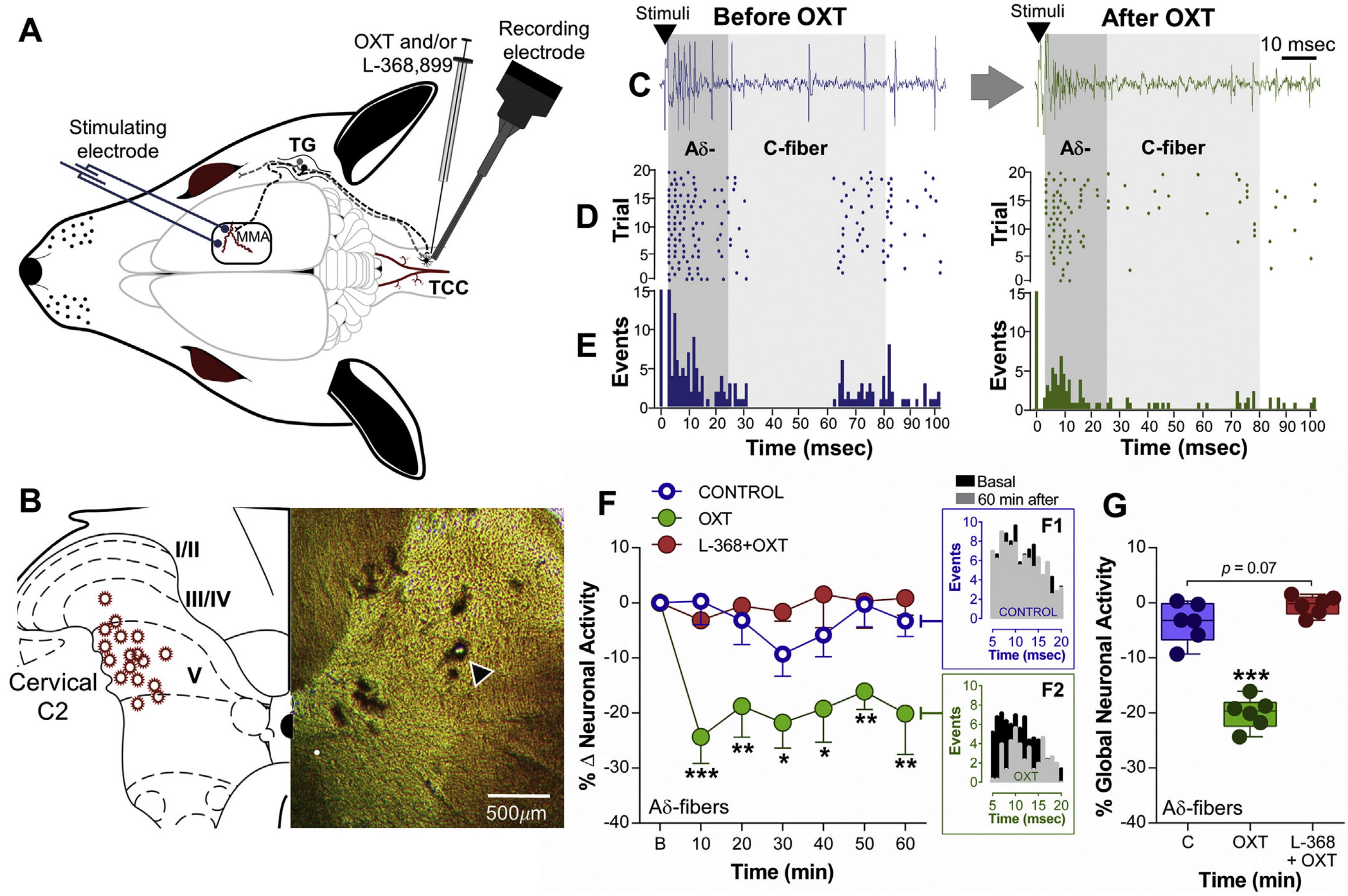

Fig. 1. Oxytocin (OXT) inhibits the nociceptive neuronal activity of wide dynamic range (WDR) cells evoked by electrical stimulation of the meninges. (A) Experimental set-up design illustrating the site of electrophysiological recording of WDR cells at TCC and the location of the electrical stimulation applied to the meninges. (B) Summary of the recording sites in the TCC region (left) and example (right) of histological microphotography for a lesion site (black arrow) in the dorsal horn at cervical C2 level. (C) Raw tracing of a single second order WDR neuron cluster response induced by one electrical stimulus on the dural region before and after OXT treatment; note that OXT diminished the neuronal activity associated with the activation of Aδ- and C-fibers. (D) Raster plot and (E) peri-stimulus time histograms (PSTH) constructed from 20 neuronal responses to dural stimulation. (F) Time course changes in the percentage average of Aδ neuronal firing induced by dural electrical stimulation and the effects of local (at TCC) OXT (20 nmol, n = 6 cells) compared with control response (n = 6 cells) or animals pre-treated with L-368,899 (20 nmol; OTR antagonist) *plus* OXT (n = 6 cells). Note that (F1 and F2) at t = 60 min, the PSTH depict the antinociceptive effect of OXT. (G) Global neuronal activity (obtained from the respective time courses) of Aδ-fibers in response to different treatments. $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$ *vs.* control.

wood-based bedding and controlled temperature (23 ± 2 °C) and humidity (50%) on a 12:12 h light/dark cycle.

2.2. Surgical procedures for electrophysiological experiments

Animals were anesthetized with 6% (*v*/v) sevoflurane delivered through a vaporizer (in a mixture of ¾ N_2O and ¼ O_2). Under these conditions, an intratracheal cannula was inserted for artificial ventilation (56 strokes/min; model 40–1002; Harvard Apparatus Ltd., Edenbridge UK) and to maintain the anesthesia. Rats were mounted on a stereotaxic frame (Kopf Instruments, Tujunga CA), and the skull was exposed to perform a craniotomy of the temporoparietal bone to gain access to the dura mater and middle meningeal artery (MMA). To access the TCC, the muscles of the dorsal neck were separated, and a cervical (C1-C2) laminectomy was performed to expose the brainstem. The dura was carefully removed. The subsequent protocols were performed under 2.5% sevoflurane. End-tidal CO_2 was monitored (Capstar-100; CWI Inc., Ardmore PA, USA), and core body temperature was maintained at 38 °C using a circulating water pad. During the experiments, the rats were paralyzed with pancuronium bromide injection (2 mg/kg; i.m.) and at the end of each experiment, animals were perfused transcardially to collect the tissue for further histological analysis (see "*Histological reconstruction of recording sites" section*).

2.3. In vivo extracellular unitary recordings at the trigeminocervical complex (TCC)

Extracellular recordings using quartz-insulated platinum/tungsten microelectrodes (4–8 MΩ) mounted in a multichannel microdrive system with integrated preamplifier (Thomas Recording GmbH, Giessen, Germany) were made from wide dynamic range (WDR) neurons activated by electrical meningeal stimulation. This WDR units also had concurrent cutaneous facial receptive field at the ophthalmic dermatome. To search for single-unit discharges, the microelectrodes were lowered (900–1400 µm from the surface) in small steps (3 µm/s). A first search of cells responding to stimulation of the ophthalmic dermatome was performed. The ophthalmic receptive field was assessed for non-noxious (brushing) and noxious (pinching) input; then, to corroborate that peripheral input relay in a WDR cell, an electrical pulse stimuli was applied (1-msec pulse duration at 0.1–3.0 mA to evoke an Aδ- and C-fiber response) by two electrodes (27 G) inserted subcutaneously into the ophthalmic receptive field. When a WDR cell was identified, it was electrically tested for a convergent input from the meninges. The test stimulation was applied using two electrodes placed on the meninges adjacent to the MMA. This test consisted of 20 square-wave stimuli at 0.5 Hz with 1-msec pulse duration at 0.1–3.0 mA to evoke an Aδ- and C-fiber response (S88 Stimulator; Grass Instruments Co., Quincy MA, USA). The peripheral stimulating electrodes were attached to a stimulus isolator unit (PSIU6; Grass Instruments Co., Warwick RI, USA).

The evoked TCC neuronal activity was amplified ×100 (1700 Differential AC amplifier; A-M Systems, Carlsborg WA, USA), digitalized, and discriminated using a CED hardware and Spike2 v5.15 software (Cambridge Electronic Design, UK). Raw and discriminated signals were fed through an audio monitor and displayed on an oscilloscope (TDS 420 A; Tektronix, Inc., Wilsonville OR, USA). Waveforms and recorded spike trains were stored on computer disk for off-line analyses. Baselines and evoked activities of the TCC neurons were recorded and analyzed as peri-stimulus time histograms (PSTHs). On this basis, the stimulating threshold to evoke action potentials and their frequency of occurrence, resulting from the stimulation of the dura mater, were attributed to the recruitment of Aδ- and C-fibers. Considering the distance between the receptive field and the recording electrode, the peak latencies observed correspond to peripheral conduction velocities within the Aδ- (3–33 msec) and C-fibers (26–80 msec). Thus, the number of action potentials elicited by the 20 receptive field stimuli was compared before (basal) and after vehicle/drug treatment.

The evoked neuronal responses were evaluated before ($t = 0$ min) and after 20 nmol oxytocin ($n = 6$ cells) (CAS No. 50–56-6; Sigma Chemical Co., St Louis, MO, USA) administration at 10, 20, 30, 40, 50 and 60 min . Furthermore, the role of OTR in the oxytocin effect was evaluated using the highly potent and selective competitive antagonist L-368,899 (20 nmol; rat p*Ki*: 8.1) (CAS No. 160312–62-9; Tocris Bioscience, Northpoint, UK), given 5 min before oxytocin ($n = 6$ cells). All drugs were carefully delivered at TCC level (topical) where the microelectrodes were positioned in a volume of 20 μl using a Hamilton® syringe. The vehicle (isotonic saline; 0.9% NaCl) and the L-368,899 did not have any effect on the nociceptive-evoked neuronal evoked activity (data not shown).

2.4. Histological reconstruction of recording sites

At the end of the electrophysiological experiments, the animals received an overdose of sevoflurane (8% *v*/v) and were transcardially perfused with 0.9% NaCl, followed by 10% formaldehyde (≈200 ml of each). Brains were post-fixed in 10% formaldehyde and cut into 40-μm serial coronal sections with a freezing microtome (Leica SM2000 R; Leica Biosystems Nussloch GmbH, Germany). The position of the electrode tip was observed with a light microscope (Fig. 1B).

2.5. Analysis

In all cases we checked that our data set followed a normal distribution (Shapiro-Wilk test) and the power of the performed test ($\alpha = 0.8$ minimum). The number of evoked potentials was normalized and expressed as percentage change from the respective baseline. This baseline was established after an identified neuron had a ≤ 10% variation in the neuronal responses induced by receptive field stimulation during 5 consecutive tests (5 min between each test). The temporal course figures are presented as mean ± standard error of the mean (S.E.M.). To evaluate the stability of the recorded neurons (only for control group) across the 60-min time frame, we used one-way repeated measures analysis of variance (RM-ANOVA); sphericity was not assumed, and corrections to degrees of freedom were made according to Greenhouse-Geisser method. Also, to compare the values of neuronal Aδ-fiber responses obtained before (basal response) and after drug treatments, we used two-way RM-ANOVA. Finally, to compare the global neuronal activity (box-whiskers plots) between different treatments a one-way ANOVA was performed. The one-way and two-way ANOVAs were followed, if applicable, by the Student-Newman-Keul's (SNK) *post hoc* test. Statistical significance was accepted at $p < .05$.

3. Results

3.1. Electrophysiological data

Extracellular unitary recordings were made from eighteen TCC neurons with convergent input from the dura mater and supraorbital receptive field (Fig. 1A). Semi-processed raw data of all neurons recorded are available at Mendeley Data (http://dx.doi.org/10.17632/65fsx78b94.1). Most recorded neurons (≈90%) were in lamina V of the TCC with a few neurons in laminae IV and VI (≈10%) at an average depth of 1095 μm (Fig. 1B). Units had an average baseline firing latency after MMA stimulation between 3 and 25 msec for Aδ-fiber component, whereas data for C-fiber responses were recorded within the range of 25–80 msec. In the case of C-fiber data, we only saw specific stimulation-linked firing in 7 units (3 in the control, 1 in the oxytocin treated, and 3 in the L-368,899 *plus* oxytocin group); thus, there was not enough power to perform a statistical analysis of these responses, an issue that it has been reported in other studies using this model (for refs. See Akerman et al., 2013). Since we recorded TCC neurons for 60-min, we first evaluated the effect of time on the neuronal responses. In the control group ($n = 6$), time had no effect on Aδ-fibers activity ($F_{2.703, 13.51} = 1.403$, $p = .284$), suggesting that no time-dependent changes in neuronal responses occurred during our experimental protocols.

3.2. Oxytocin inhibits the nociceptive-evoked responses by electrical dural stimulation in WDR neurons by OTR activation

Fig. 1C-1E shows an example of the electrophysiological data obtained as raw tracing data from one WDR cell and the raster plot and the PSTH before and after oxytocin treatment. It is interesting to note that oxytocin diminishes the evoked neuronal activity associated with the activation of Aδ- and C-fibers. Fig. 1F shows the formal analysis of Aδ-fiber responses induced by electrical meningeal stimulation in control cells ($n = 6$) and cells pre-treated with oxytocin (20 nmol, n = 6 cells) and/or L-368,899 (20 nmol, n = 6 cells).

The two-way RM ANOVA revealed that oxytocin treatment statistically diminishes the dural induced-neuronal activity of TCC WDR neurons ($F_{2, 15} = 9.078$, $p = .003$) in a time dependent manner ($F_{6, 90} = 3.926$, $p = .002$). The inhibitory effect of oxytocin started at 10 min and lasted up to 60 min as revealed by the SNK *post hoc* test. Similarly, analyzing these data as a global neuronal firing (Fig. 1G), we found a strong effect of oxytocin on the dural nociceptive Aδ-fiber input ($F_{2, 15} = 84.046$, $p \leq .001$).

To investigate if OTR was involved in the oxytocin-induced antinociception, a potent and selective antagonist to OTR was used. Pretreatment with L-368,899 (20 nmol) abolished the oxytocin-induced antinociception. Furthermore, there are no differences between the responses obtained in the control group *vs.* L-368,899 *plus* oxytocin (Fig. 1F and 1G).

4. Discussion

4.1. General

Our experiments using electrophysiological unitary recordings of second order neurons in the TCC, suggest that the nociceptive input processed at TCC and arriving from the activation of the trigeminovascular system could be blocked by oxytocin-dependent mechanisms. Moreover, the oxytocin-induced antinociception may be attributable to OTR activation, thus explaining the probable mechanisms involved in the antimigraine effects observed in some studies in humans (Tzabazis et al., 2016, 2017; Wang et al., 2013; You et al., 2017). These data provide a preclinical proof of concept about the potential role of OTRs modulating nociceptive transmission relevant to migraine pathophysiology.

4.2. Oxytocin administration inhibits the dural electrically-induced neuronal firing of TCC

Some short clinical reports in humans suggest that oxytocin, a hypothalamic neuropeptide, could act as an antimigraine molecule when given by intravenous (Phillips et al., 2006) or intranasal (Tzabazis et al., 2017) route; however, it remains unknown how this hypothalamic neuropeptide could be acting as an antimigraine agent. From a physiological perspective, functional imaging and deep brain stimulation studies point to the hypothalamus as a key player in the pathophysiology of migraine (Denuelle et al., 2007). Most specifically, the paraventricular nucleus of the hypothalamus (PVN) has been implicated, as it not only sends neuronal projections to the Sp5C (a region that receives input from the meningeal Aδ- and C-fiber nociceptors) (Abdallah et al., 2013; Robert et al., 2013) but also controls the neuronal activity of Sp5C neurons (Robert et al., 2013). In this context, our results showed that: (i) peripheral electrical stimulation of dural tissue induces nociceptive responses at TCC level; (ii) spinal topical (at TCC level) administration of oxytocin can specifically inhibit the electrically-induced nociceptive responses of nociceptive-specific WDR neurons; and (iii) pre-treatment with L-368,899 (a selective and potent OTR antagonist) abolished the above inhibition by oxytocin. These findings using trigeminovascular electrical activation as a migraine model suggest that oxytocinergic mechanisms at TCC level may play a role in the modulation of nociceptive input.

Certainly, trigeminovascular activation is believed to be involved in the pathophysiology of migraine (Akerman et al., 2013), and electrical stimulation of meningeal structures is painful in humans. Accordingly, electrical meningeal stimuli activate nociceptive neurons located at TCC level, and compounds that inhibit this trigeminovascular dural nociception have been shown to be predictive of efficacy against migraine (for refs. See Akerman et al., 2013). Since oxytocin inhibits the firing of dural-evoked nociceptive second-order trigeminovascular neurons arriving at TCC, our findings reveal a potential involvement of oxytocin in migraine pathophysiology. Furthermore, binding sites for oxytocin have been described at TCC level, and OTRs have been reported at the trigeminal ganglion (Tzabazis et al., 2016, 2017). Also, similar data obtained in an electrophysiological model of orofacial pain (García-Boll et al., 2018) tend to support our contention that oxytocin inhibits the nociceptive activation of TCC neurons.

4.3. A role for OTR in the oxytocin-induced inhibition of trigeminovascular evoked responses

A further aim was to investigate whether OTR is involved in the oxytocin-induced antinociception at TCC, by administering a potent and selective OTR antagonist (L-368,899; rat pK_i: 8.1 with 50-fold selectivity over vasopressin V_{1A} receptors [V_{1A}R]). Our data showed that pre-treatment with L-368,899 abolishes the effect induced by oxytocin. The simplest interpretation of this result implies that oxytocin at TCC level interacts with OTRs (probably located in the primary nociceptive afferent fibers) blocking the nociceptive neuronal transmission arriving from the MMA. The precise molecular mechanism has not been well described but an *in vitro* study by Gong et al. (2015) showed that oxytocin induces hyperpolarization of primary afferent fibers by the OTR/Ca^{2+}/nNOS/NO pathway. Accordingly, several lines of behavioral and electrophysiological evidence have shown that oxytocin inhibits nociception at spinal lumbar level by OTR mechanisms (for refs. See González-Hernández and Charlet, 2018). Most importantly, in human tissue treated with SR-49059 (an antagonist to V_{1A}R), Freeman et al. (2017) showed that ^{125}I-ornitine vasotocin (an oxytocin radioligand) has specific binding sites at the spinal trigeminal nuclei, suggesting that oxytocin could act on OTR at this level.

4.4. Final considerations and conclusion

Although more physiologically models are necessary to evaluate the potential antimigraine effect of oxytocin, we would like to emphasize that the approach followed to test the relevance of oxytocinergic transmission modulating trigeminovascular dural nociception at TCC is a predictive model to test potential antimigraine molecules (Akerman et al., 2013). In any case, confirmatory studies evaluating the oxytocin effect in a more integrative approach using conscious behavioral assays will be relevant.

Finally, although a vast array of drugs is available to treat migraine (acute and prophylactic), searching for new molecules against migraine is still imperative and in the last fifteen years, oxytocin has emerged as an analgesic molecule with action at peripheral, spinal and supraspinal levels and more recently the effect of this peptide against migraine pain has been suggested. Intranasal administration of this peptide has been proposed (Wang et al., 2013; Tzabazis et al., 2017) considering that by this route oxytocin could reach the TCC (Tzabazis et al., 2017). Within this framework, our work provides evidence supporting the notion that oxytocin by OTR activation inhibits the nociceptive input of dural-evoked nociception at TCC, and, it reveals a plausible target for oxytocin in migraine treatment.

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Author contributions

EGB and GML conducted and collected experimental data. AGH and MCL conceived, designed and supervised the experiments. EGB and AGH analyzed experimental data and wrote the manuscript. All authors critically read and edited the manuscript.

Declaration of Competing Interest

The authors declare that there is no conflict of interest.

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